AI Assisted Coding

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Batch : 34

Course Code : 23CS002PC304

Assignment Number : 15.5

Lab 15 – Backend API Development: Creating RESTful services with AI

Task 1 – Student Attendance API

Task:

Create a RESTful API for student attendance tracking.

Instructions:

• Endpoints required:

o GET /attendance → View attendance records

o POST /attendance → Mark attendance

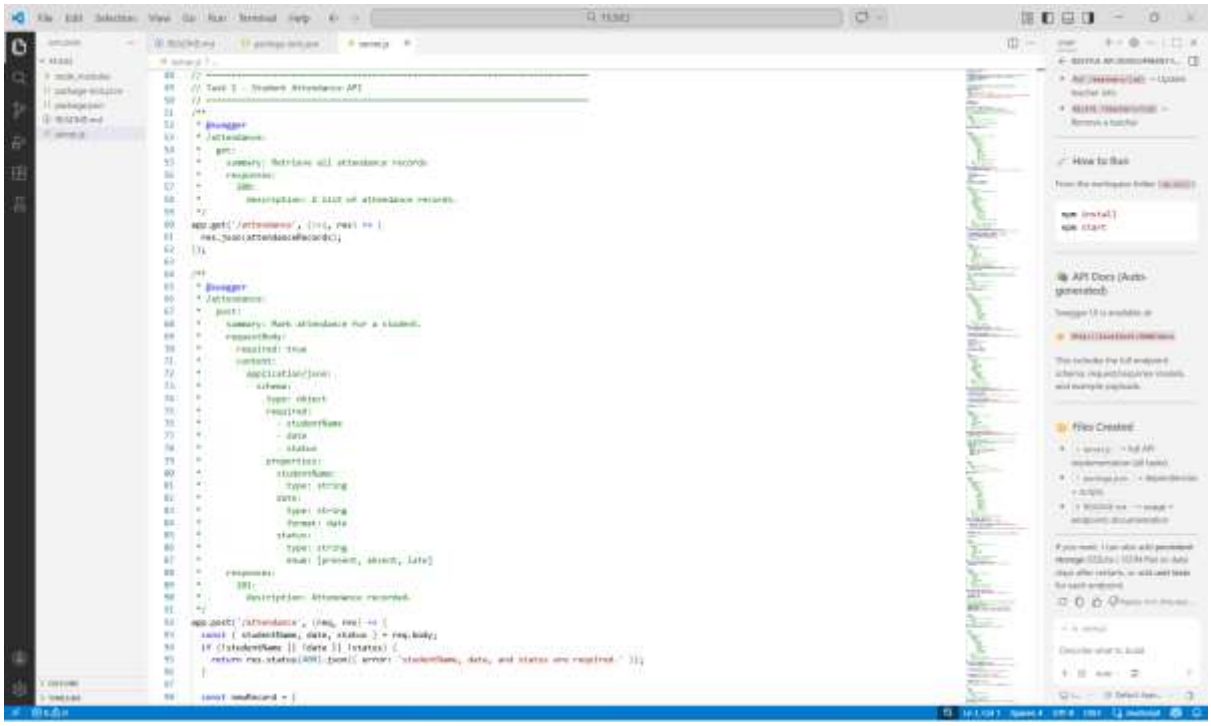
o PUT /attendance/{id} → Update attendance record

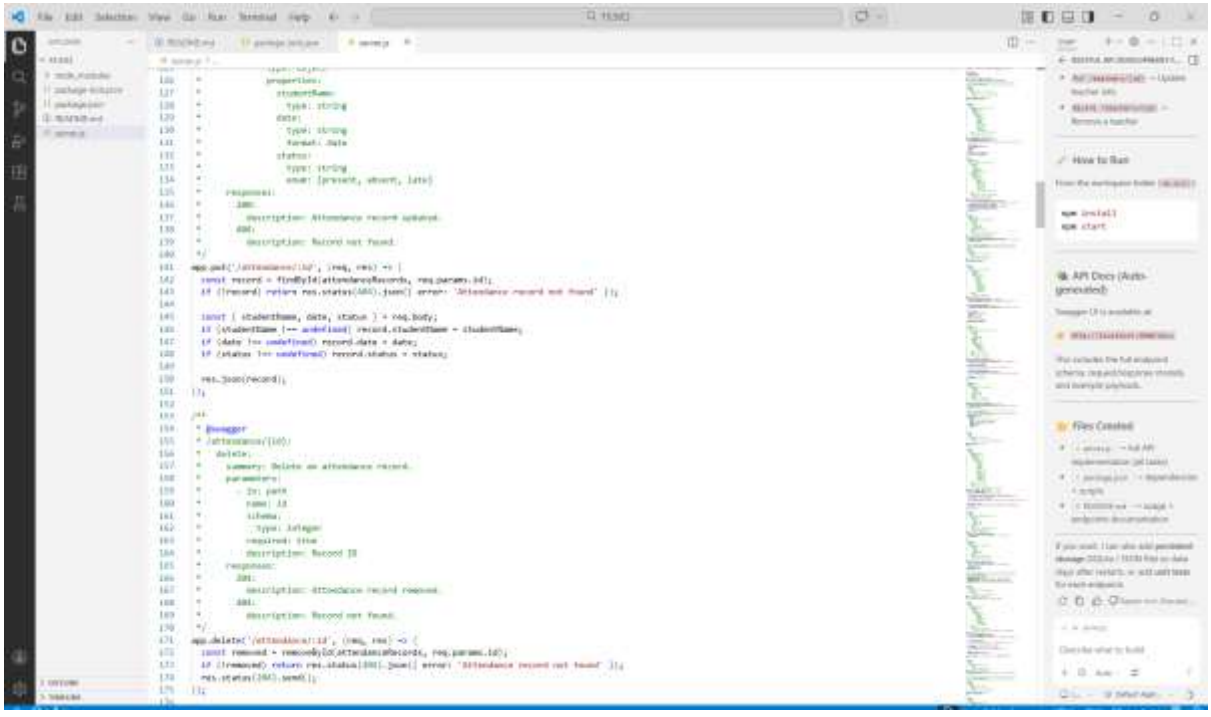
o DELETE /attendance/{id} → Remove attendance entry

• Ensure JSON-based request and response handling.

Expected Output:

• API for maintaining and updating attendance records.





Task 2 – Inventory Management API

Task:

Develop a REST API for warehouse inventory management.

Instructions:

• Endpoints required:

o GET /inventory → List all items

o POST /inventory → Add new item

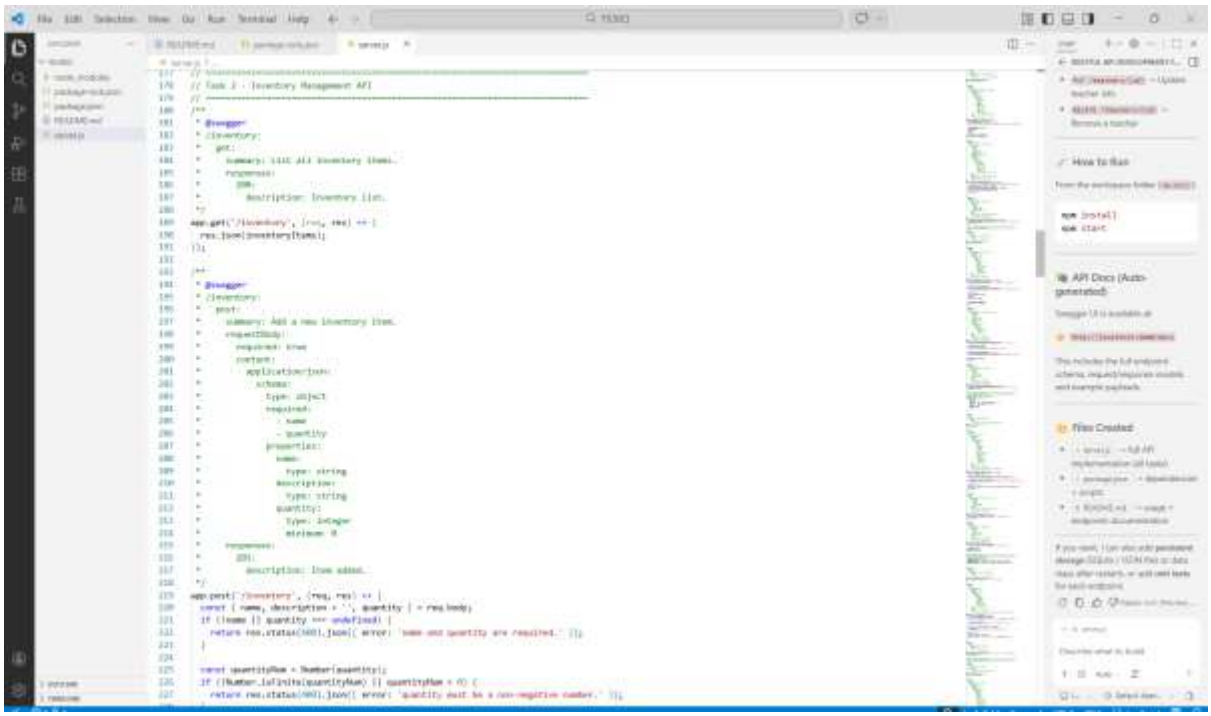
o PUT /inventory/{id} → Update stock quantity

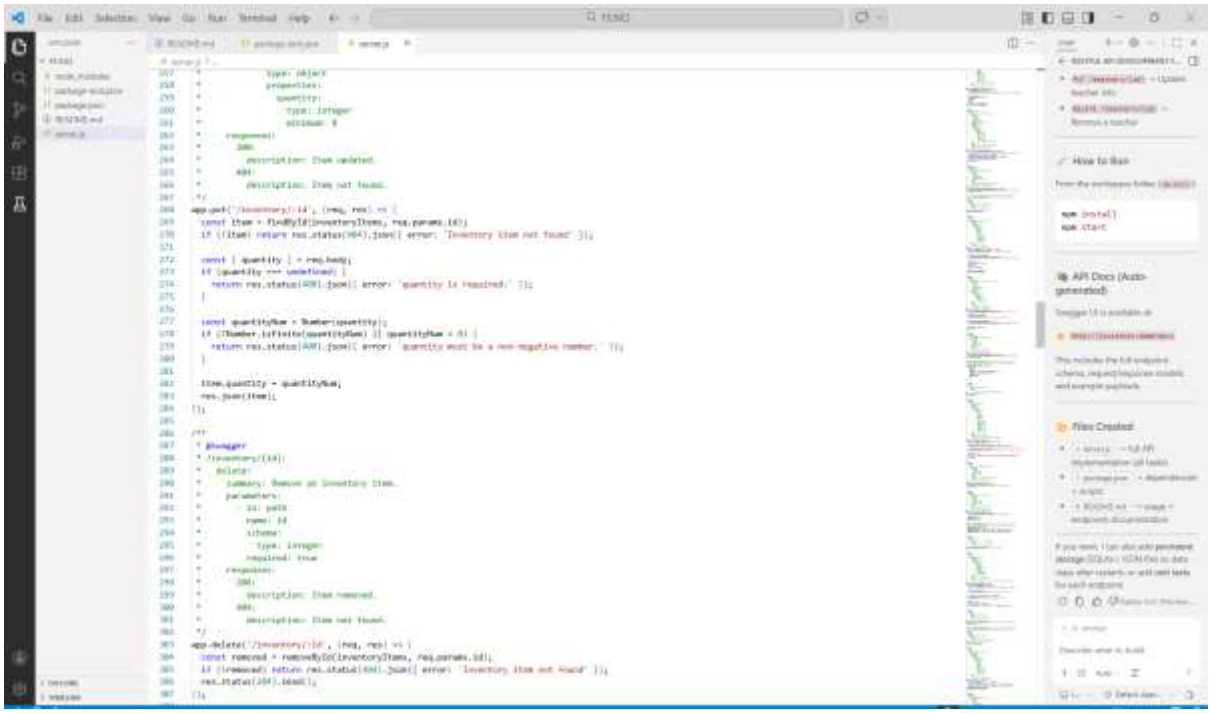
o DELETE /inventory/{id} → Remove an item

• Implement validation for stock values.

Expected Output:

• API capable of managing inventory stock levels.





Task 3 – Hotel Room Booking API

Task:

Develop a RESTful API for hotel room booking.

Instructions:

• Endpoints required:

o GET /rooms → View available rooms

o POST /rooms/book → Book a room

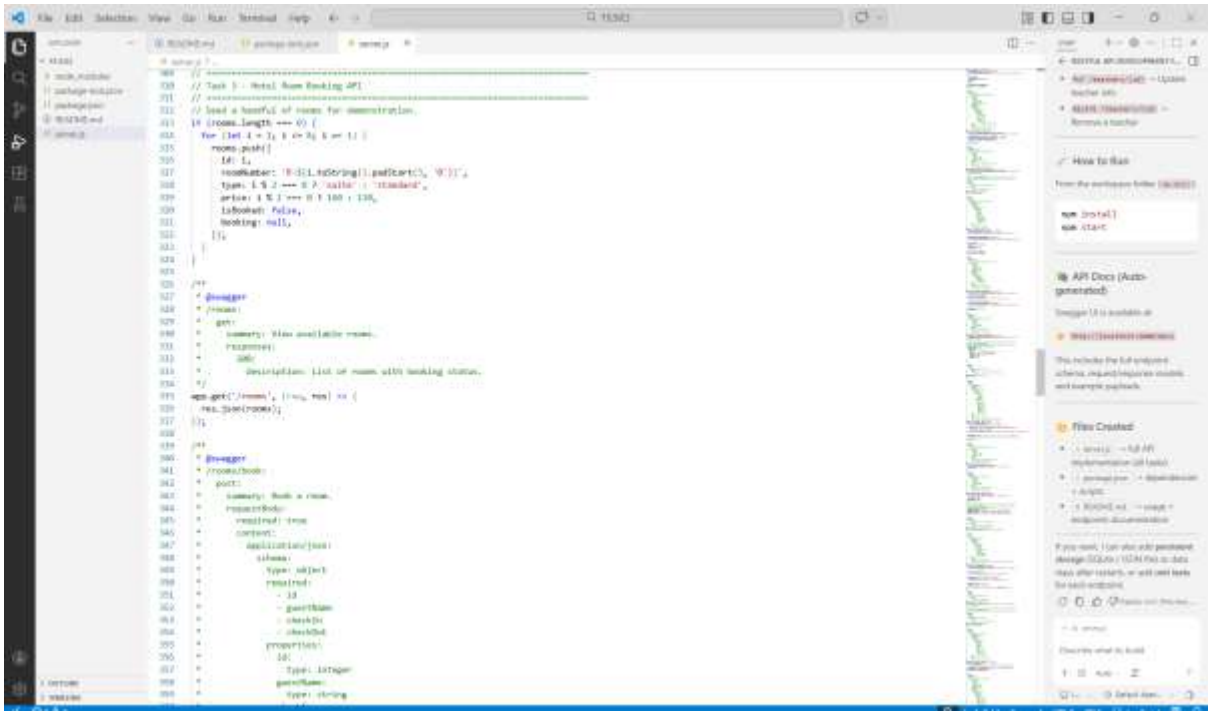
o GET /rooms/{id} → View booking details

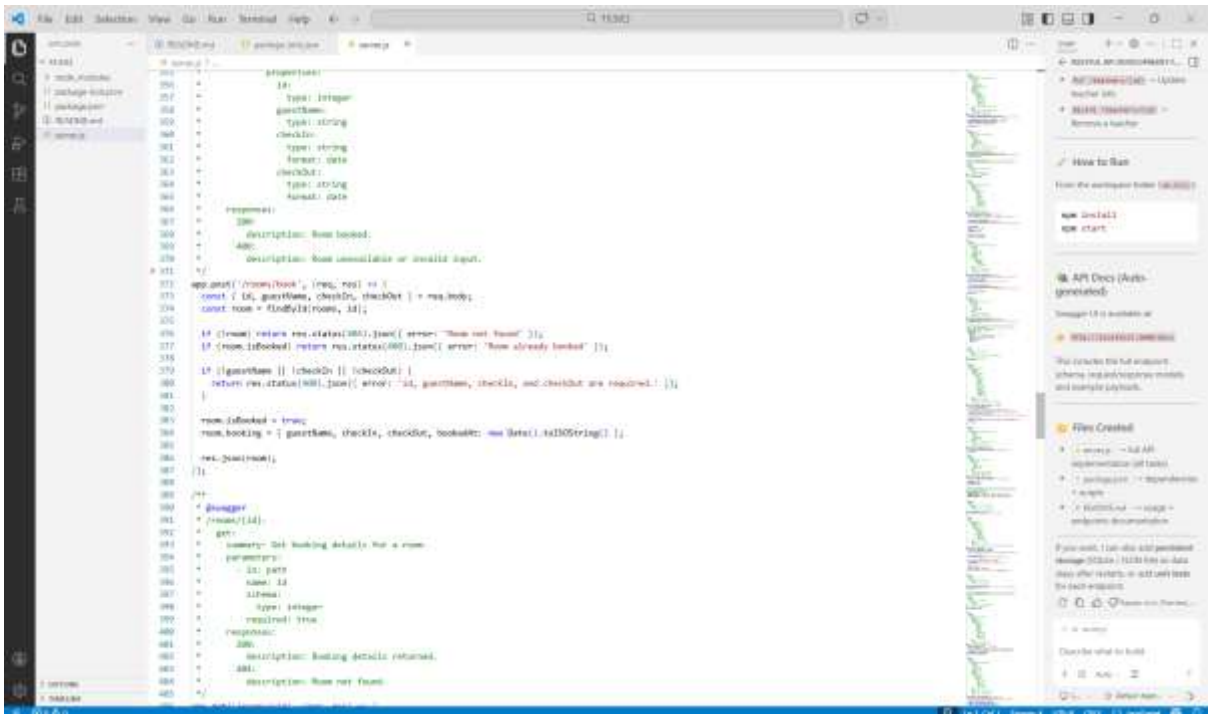
o DELETE /rooms/{id} → Cancel booking

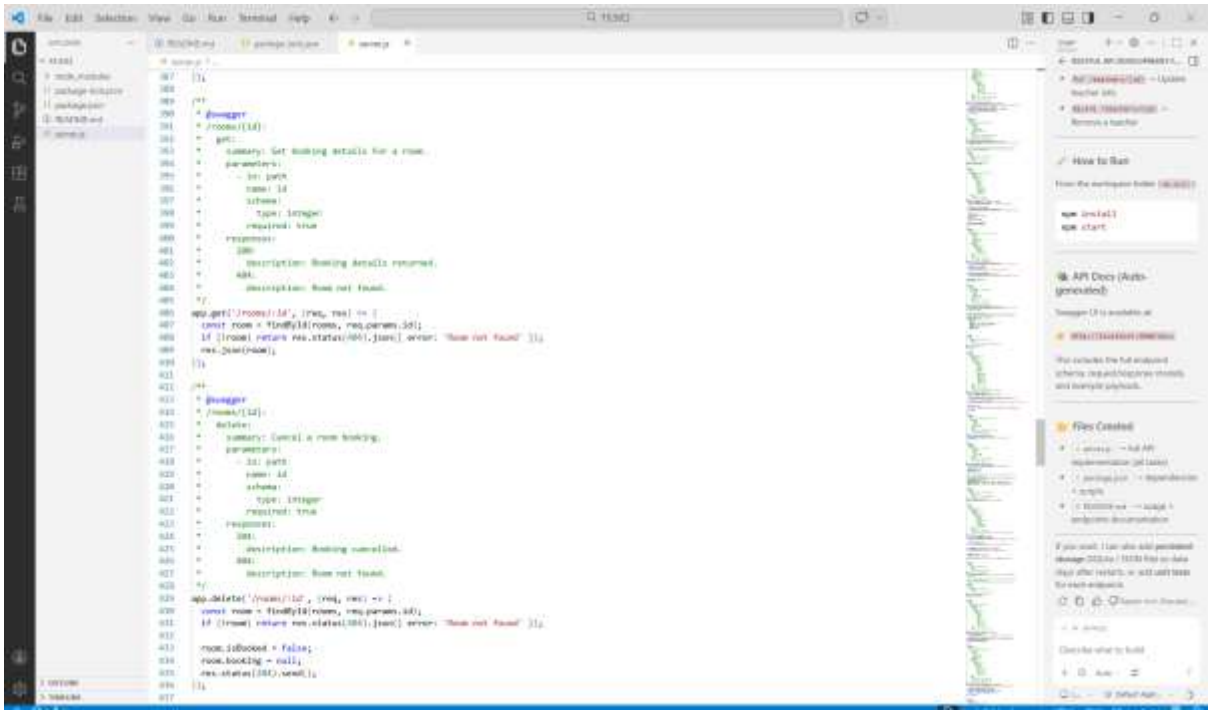
• Implement error handling for unavailable rooms.

Expected Output:

• API simulating hotel room booking operations.







Task 4 – Online Voting API

Task:

Create a RESTful API for an online voting system.

Instructions:

• Endpoints required:

o GET /candidates → List candidates

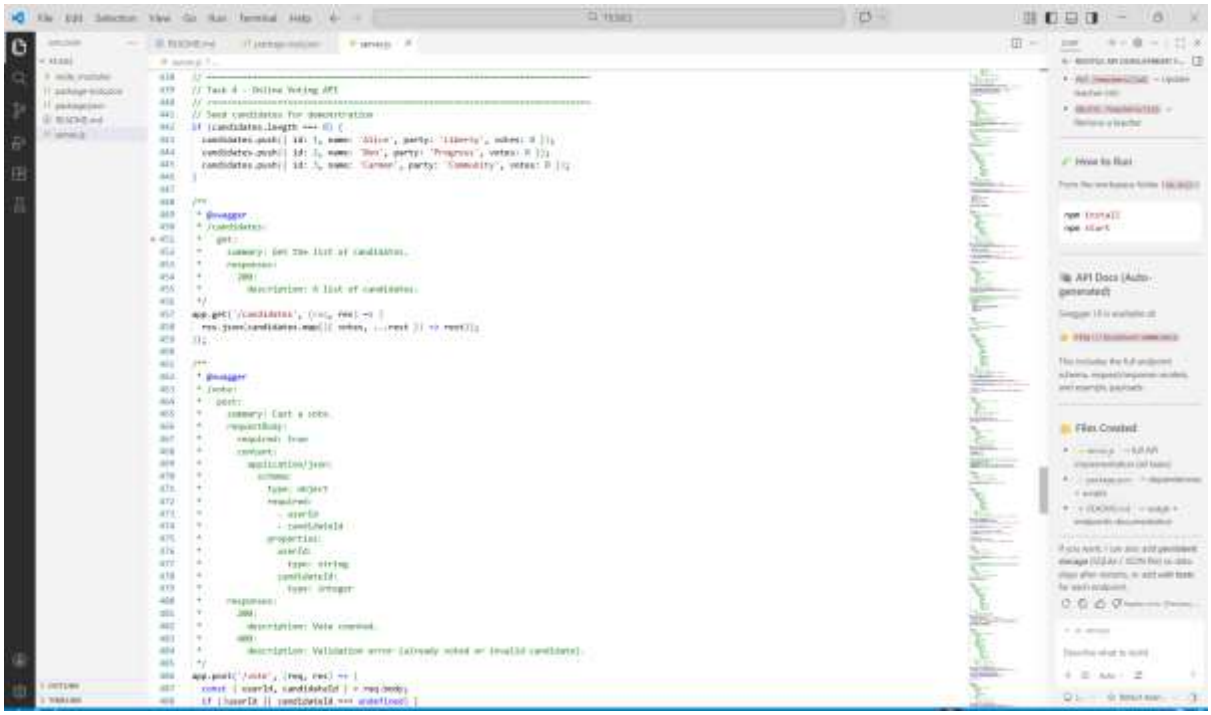
o POST /vote → Cast a vote

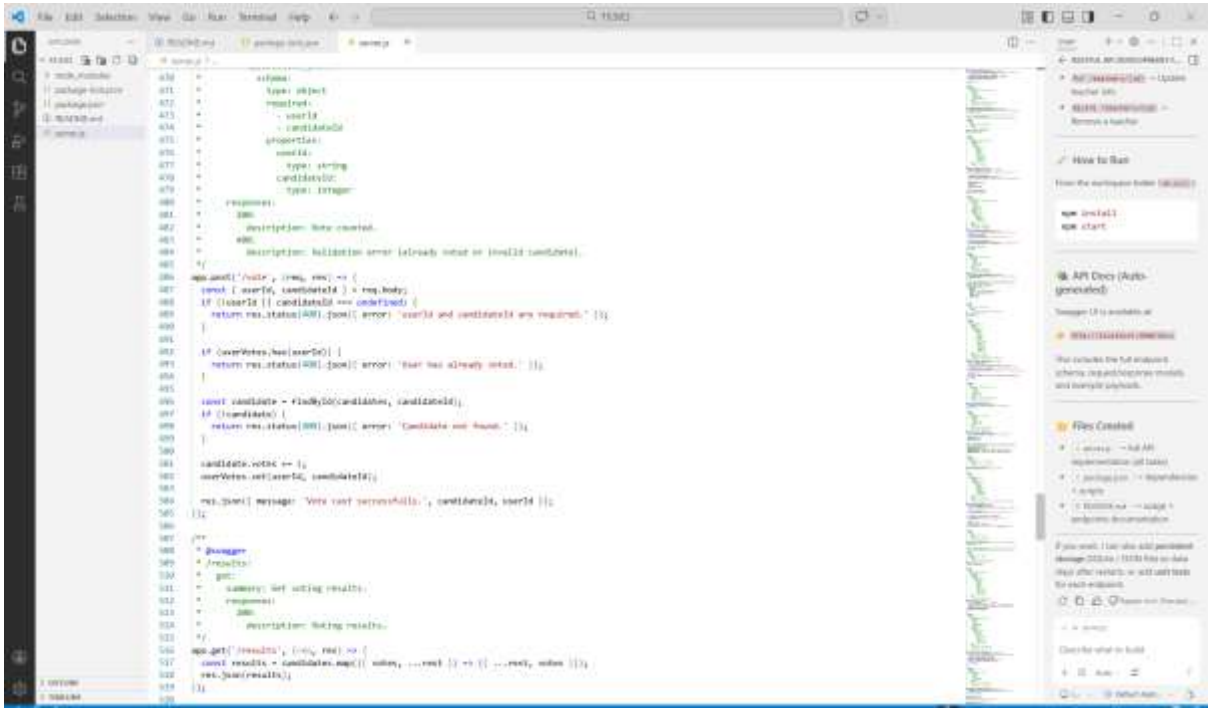
o GET /results → View voting results

• Ensure one vote per user using in-memory validation.

Expected Output:

• API simulating a secure online voting process.





Task 5 – Teacher Management API

Task:

Develop a RESTful API to manage teacher records.

Instructions:

• Endpoints required:

o GET /teachers → List all teachers

o POST /teachers → Add a new teacher

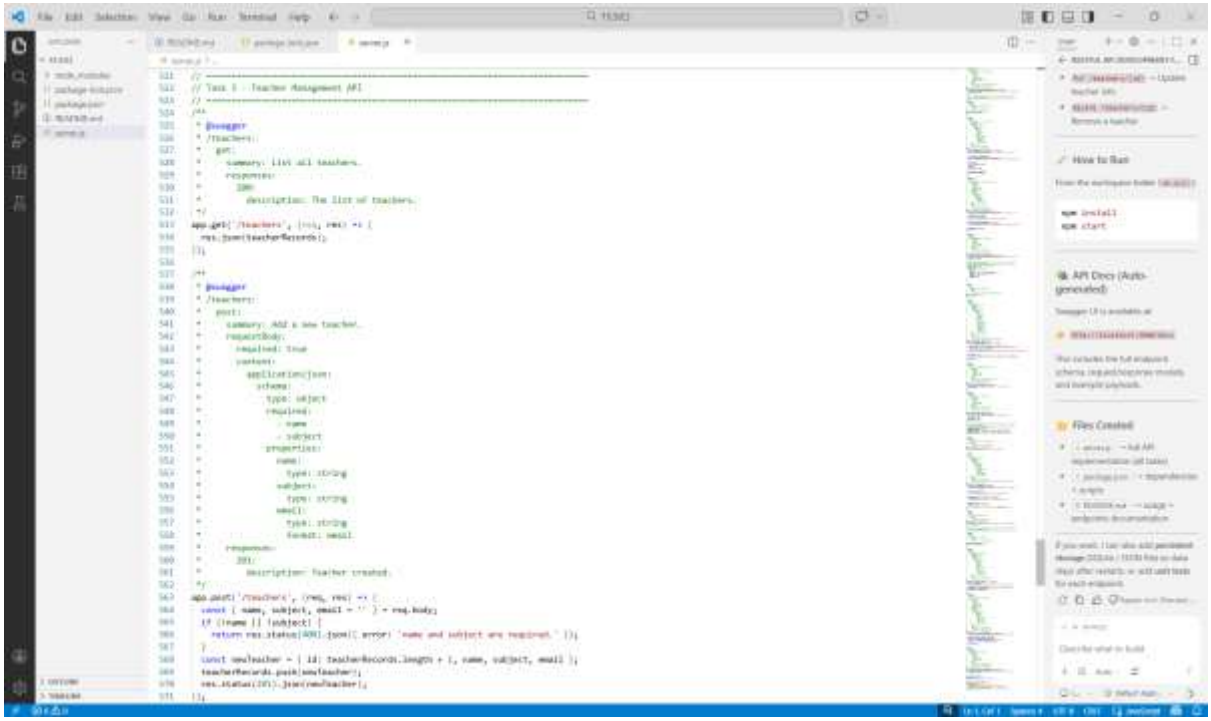
o PUT /teachers/{id} → Update teacher details

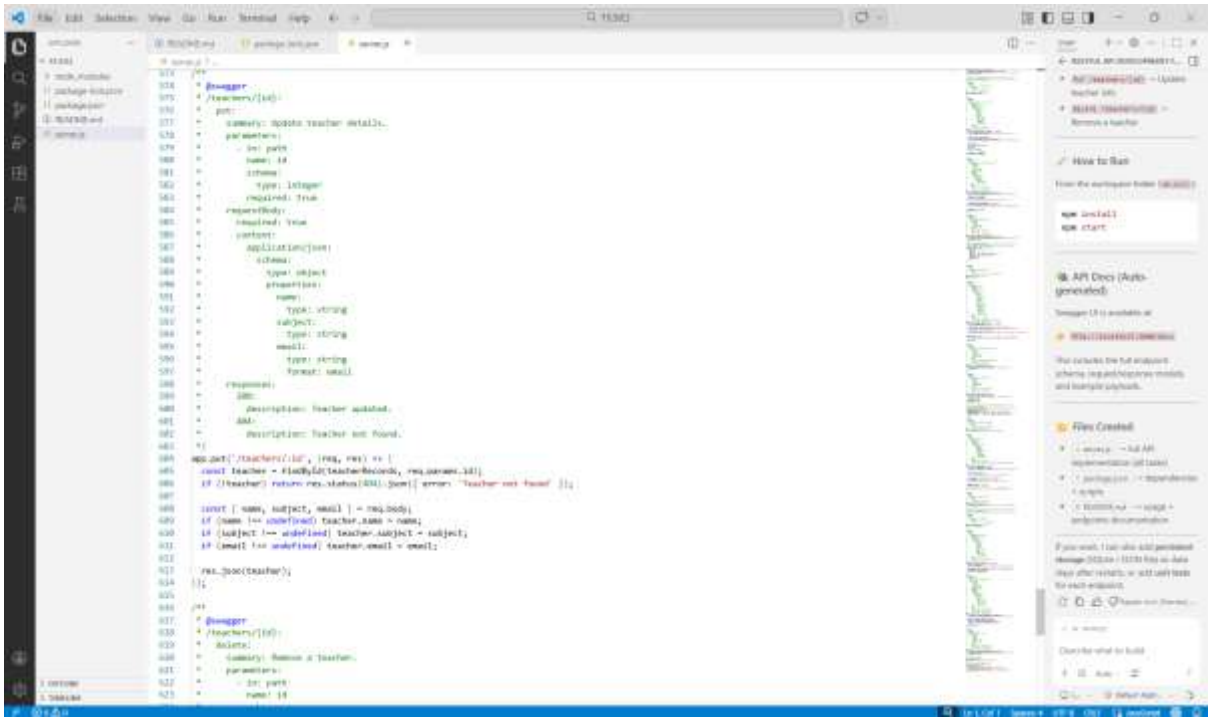
o DELETE /teachers/{id} → Remove a teacher

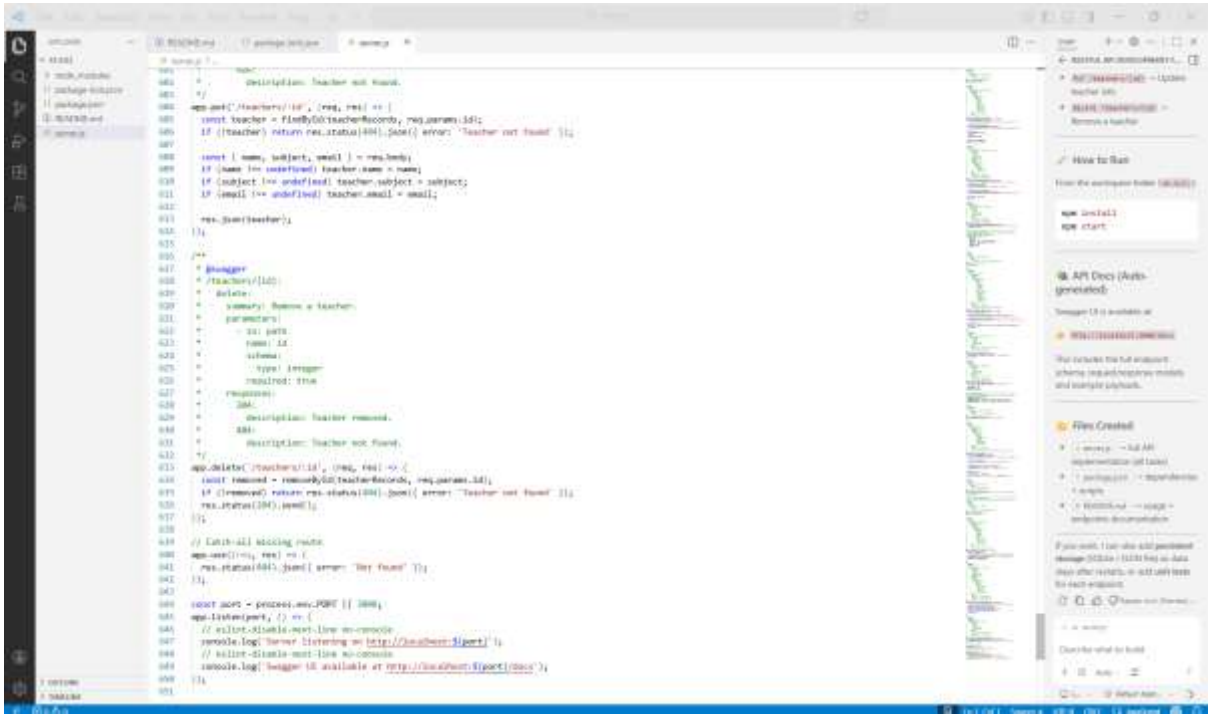
• Ensure proper error handling and JSON responses.

Expected Output:

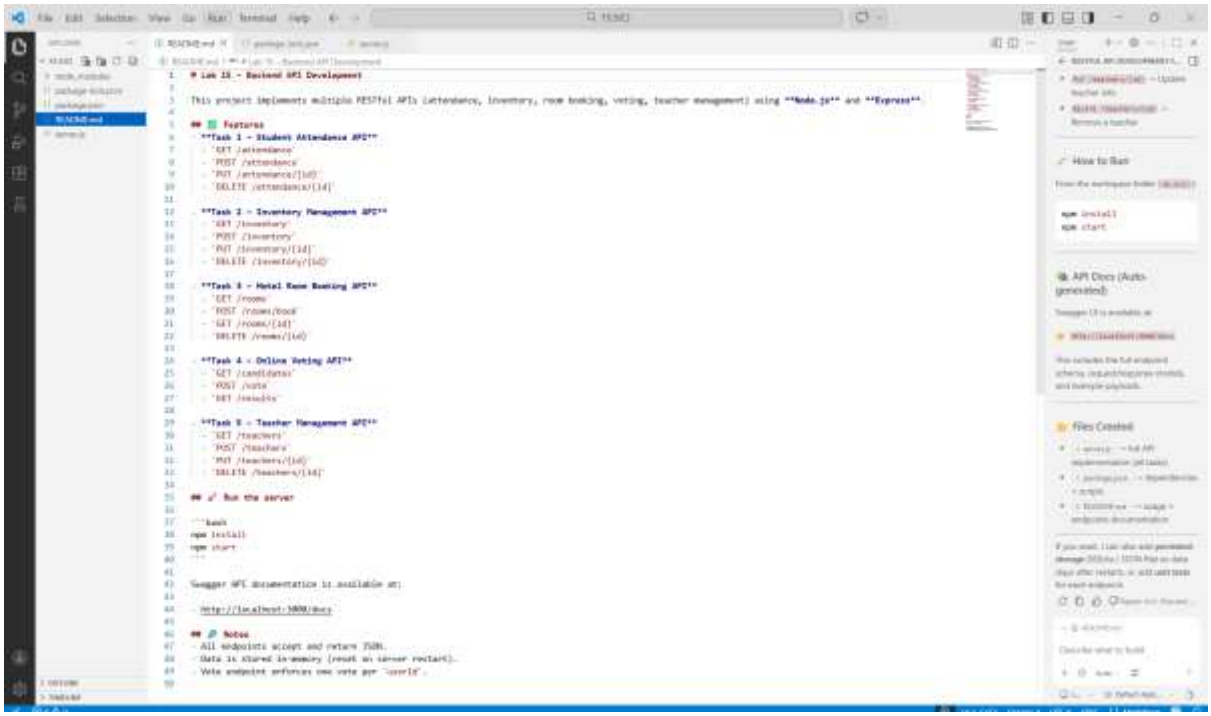
• Functional API for managing teacher information.



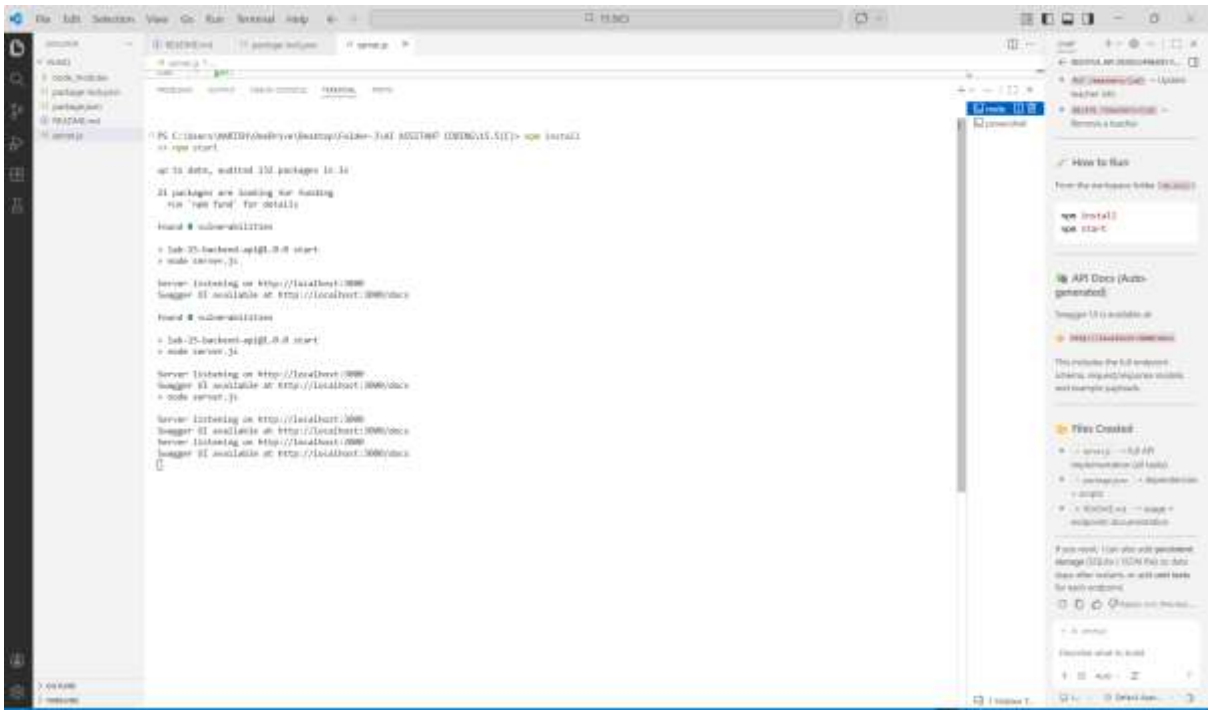




Backend API Development



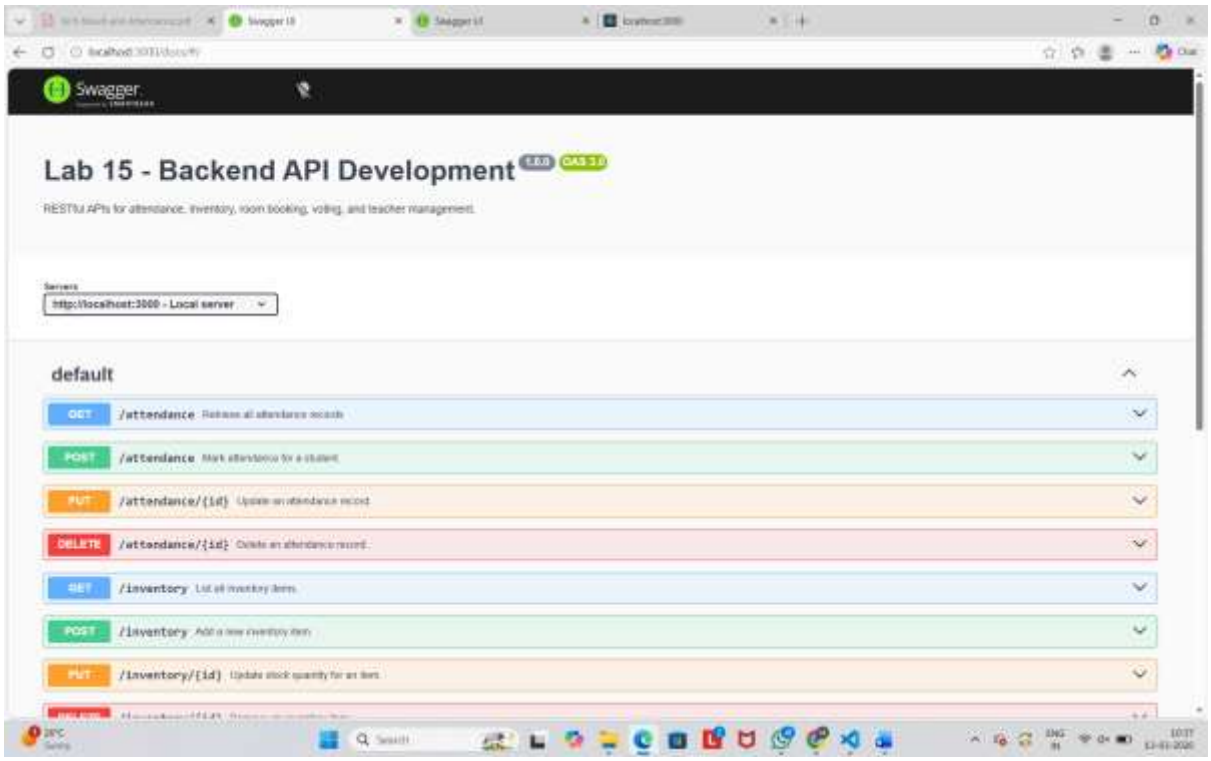
Install Project & Run the Project

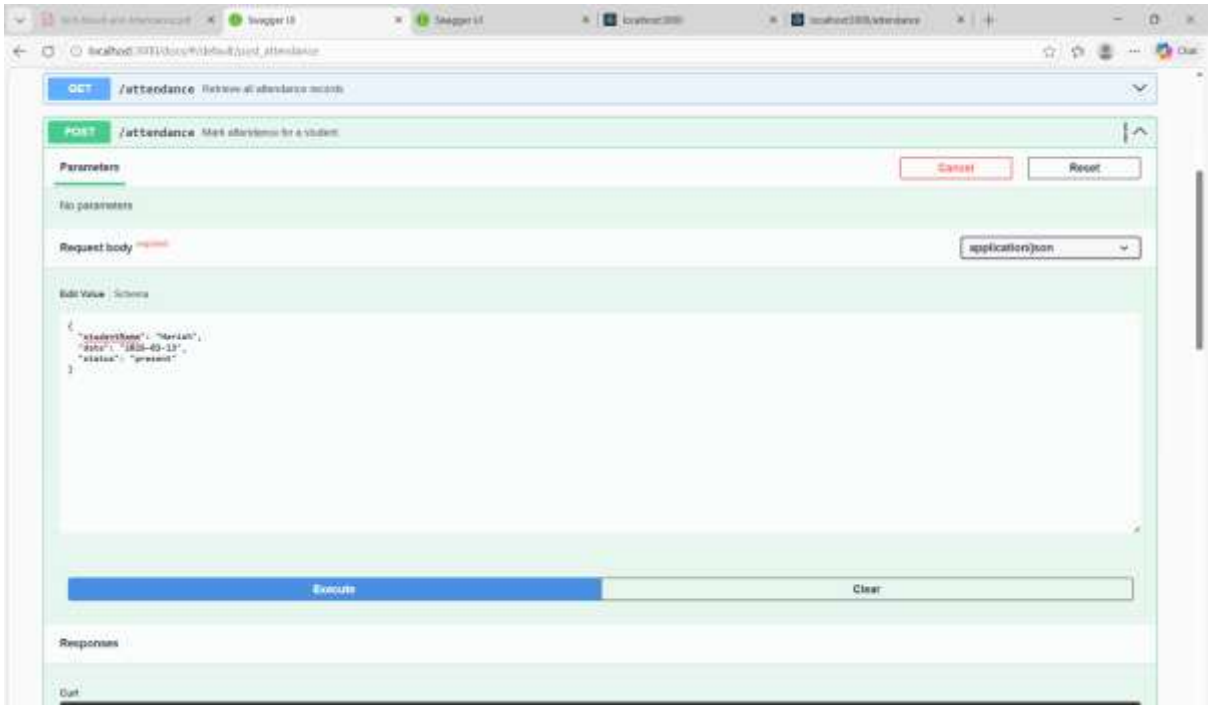


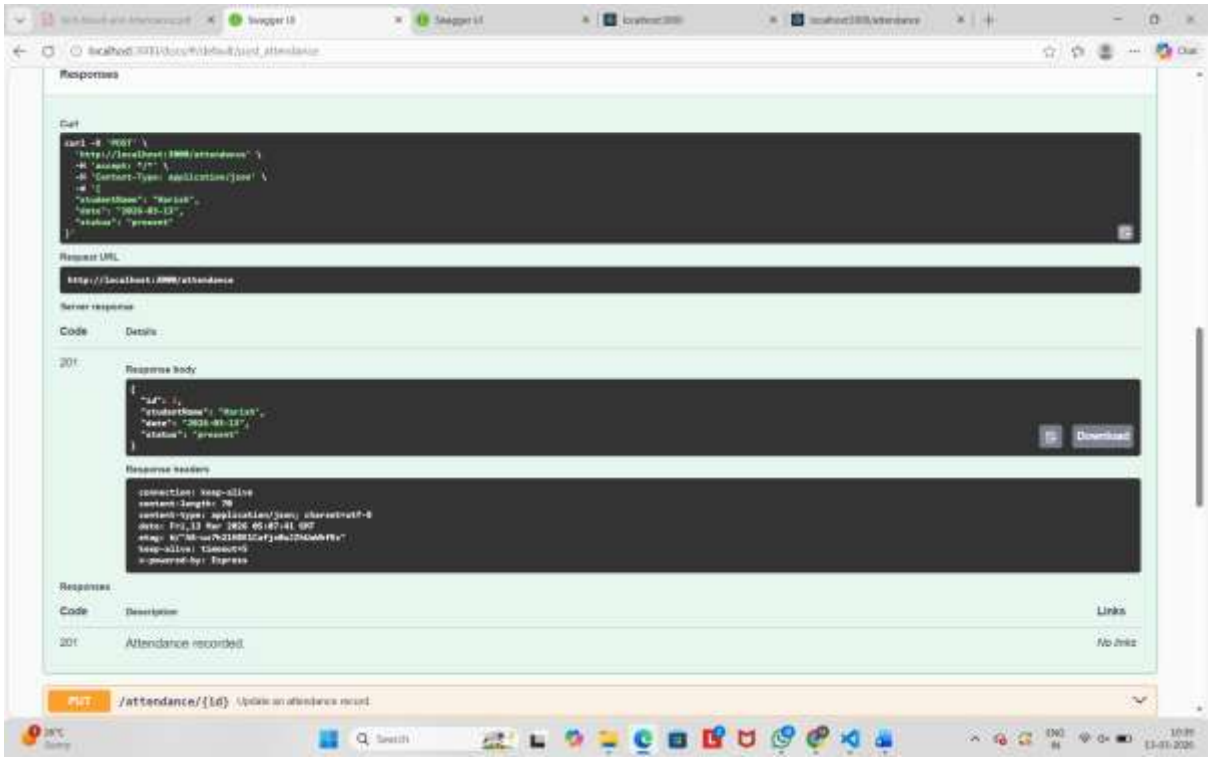
Server listening on http://localhost:3000

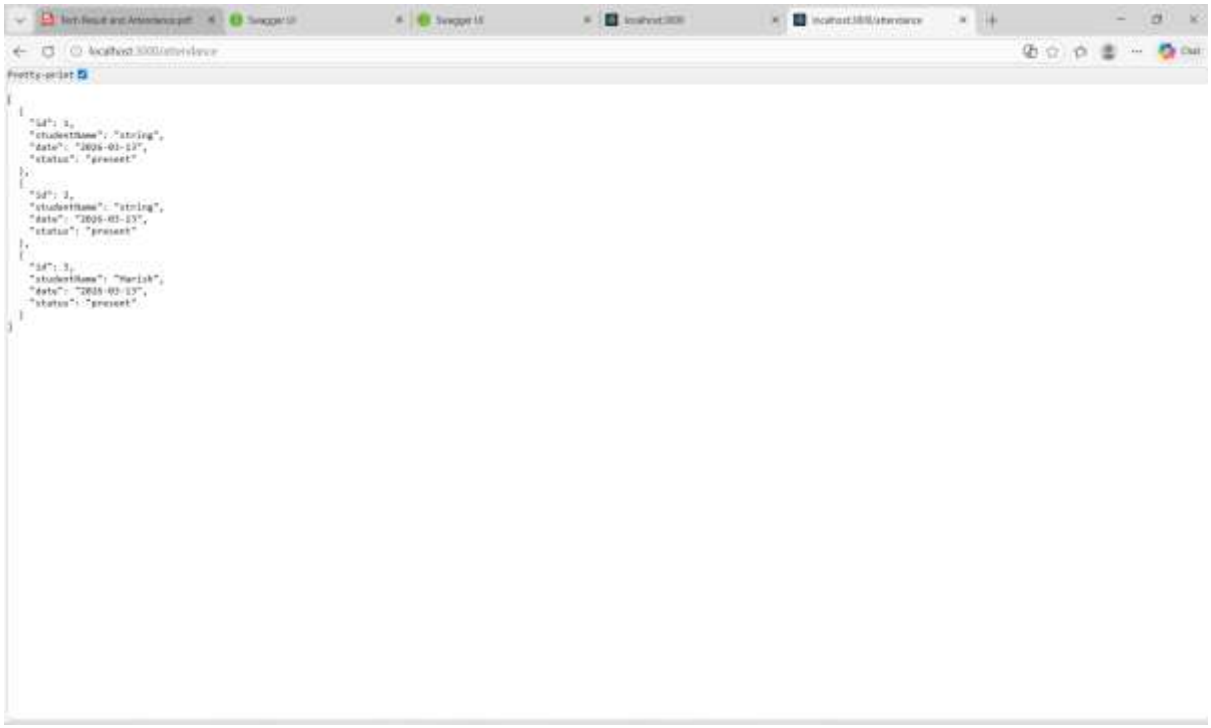
Swagger UI available at http://localhost:3000/docs

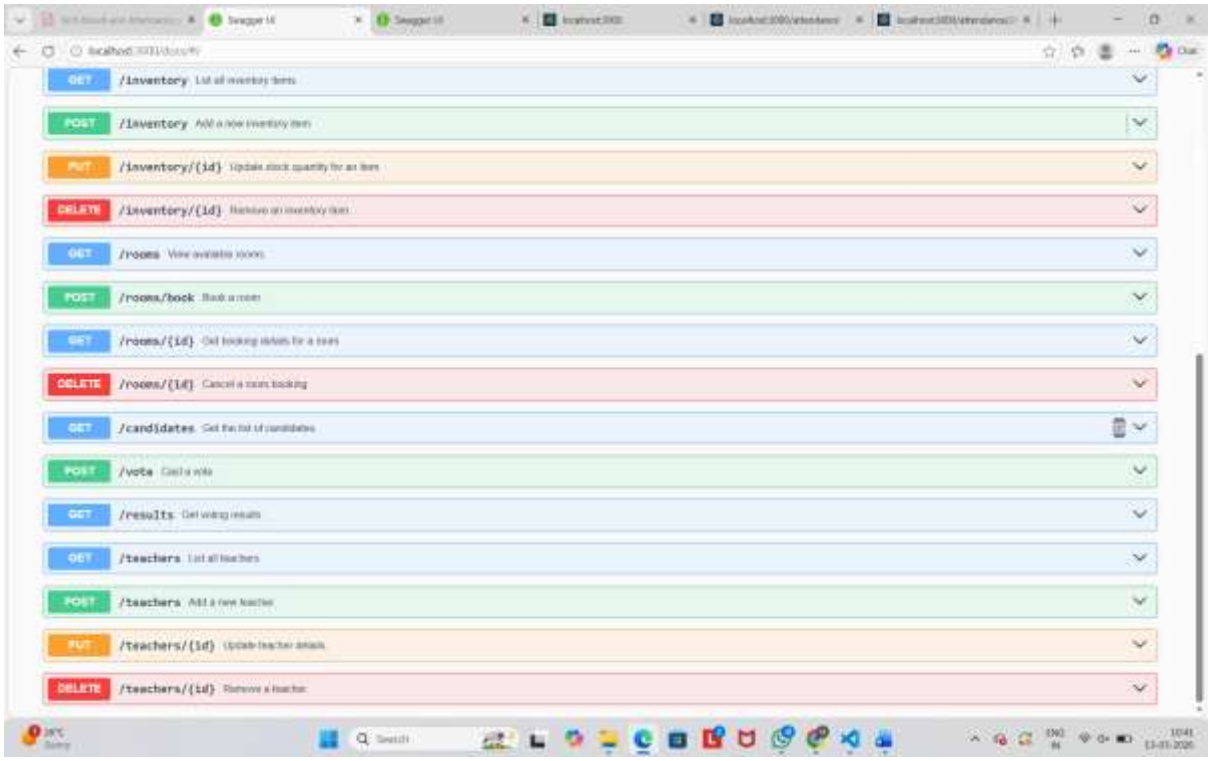
http://localhost:3000/docs :

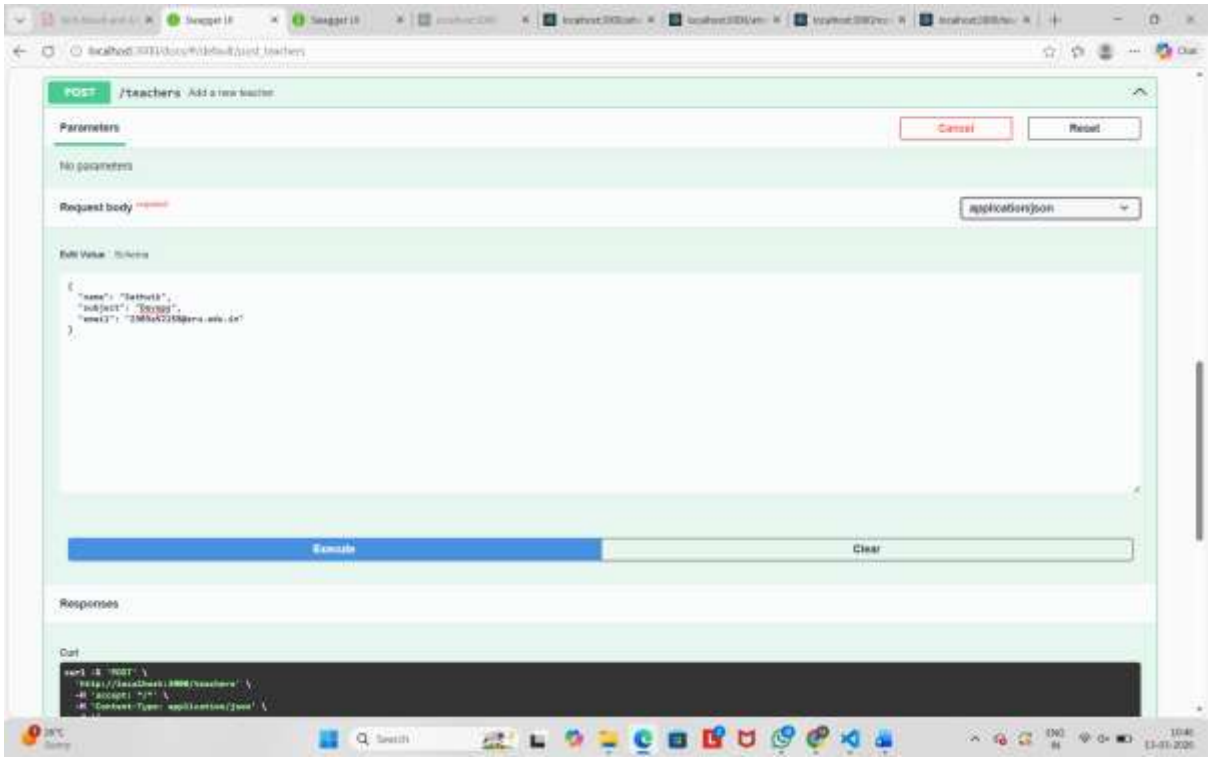

Swagger
Lab 15 - Backend API Development
RESTful APIs for attendance, inventory, room booking, voting, and teacher management.
Servers
http://localhost:3000 - Local server
default
GET /attendance
POST /attendance
PUT /attendance/{id}
DELETE /attendance/{id}
GET /inventory
POST /inventory
PUT /inventory/{id}

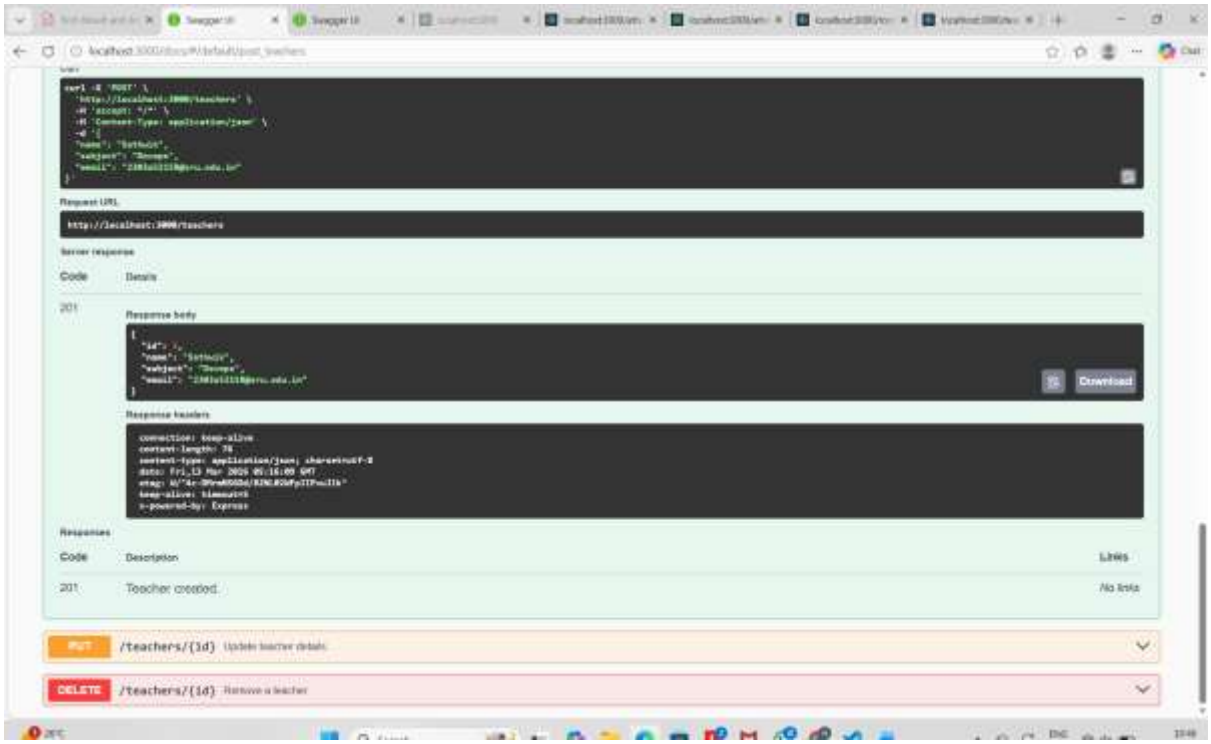

GET /attendance
POST /attendance
Parameters
Cancel
Reset
No parameters
Request body
application/json
Edit Value
Schema
Execute
Clear
Responses
Curl

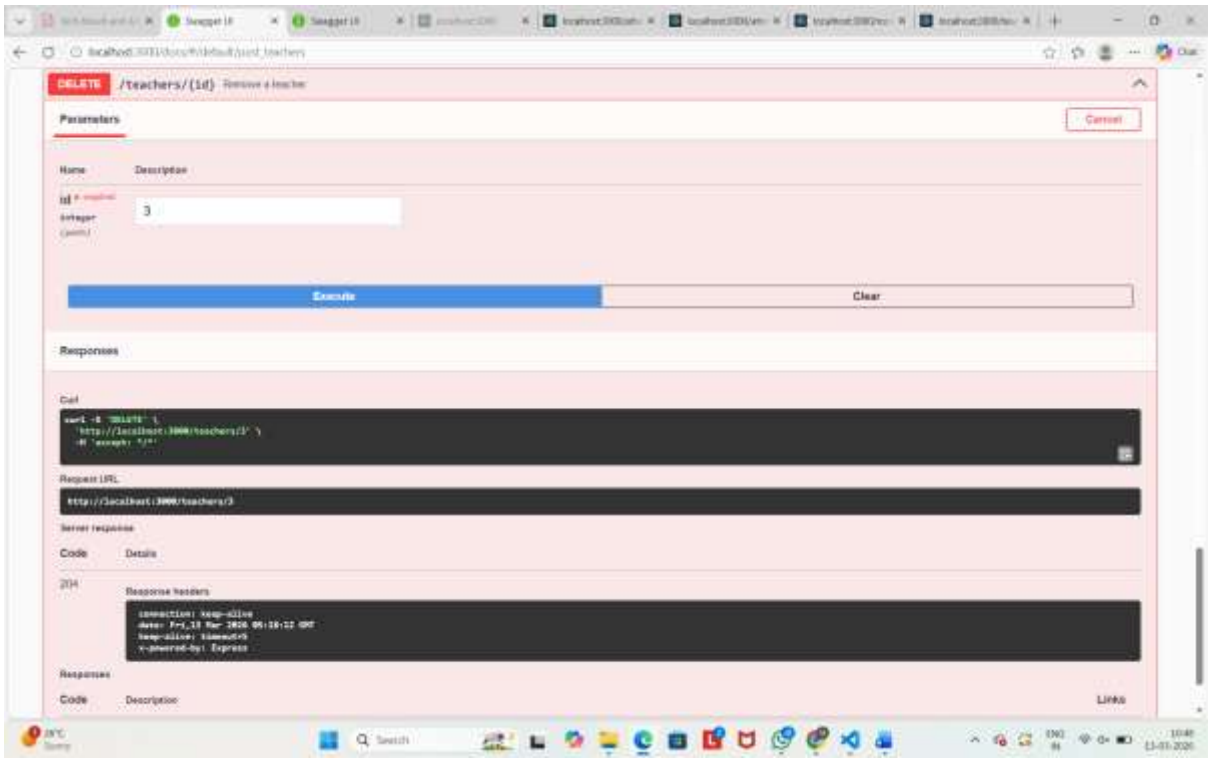

Responses
Curl
curl -X 'POST' \
'http://localhost:8000/attendance' \
-H 'accept: */*' \
-H 'Content-Type: application/json' \
-d '{
"studentName": "Harish",
"date": "2026-03-13",
"status": "present"
}'
Request URL
http://localhost:8000/attendance
Server response
Code
Details
201
Response body
{
"id": 1,
"studentName": "Harish",
"date": "2026-03-13",
"status": "present"
}
Download
Response headers
connection: keep-alive
content-length: 70
content-type: application/json; charset=utf-8
date: Fri,13 Mar 2026 05:07:41 GMT
etag: W/"46-wn/9s2lHBSlCafjnRuI2hGAbRfk"
keep-alive: timeout=5
x-powered-by: Express
Responses
Code
Description
Links
201
Attendance recorded
No links
PUT
/attendance/{id}
Update an attendance record


Pretty-print
{
"id": 1,
"studentName": "string",
"date": "2026-03-13",
"status": "present"
},
{
"id": 2,
"studentName": "string",
"date": "2026-03-13",
"status": "present"
},
{
"id": 3,
"studentName": "Harish",
"date": "2026-03-13",
"status": "present"
}
]


GET /inventory List all inventory items
POST /inventory Add a new inventory item
PUT /inventory/{id} Update stock quantity for an item
DELETE /inventory/{id} Remove an inventory item
GET /rooms View available rooms
POST /rooms/book Book a room
GET /rooms/{id} Get booking details for a room
DELETE /rooms/{id} Cancel a room booking
GET /candidates Get the list of candidates
POST /vote Cast a vote
GET /results Get voting results
GET /teachers List all teachers
POST /teachers Add a new teacher
PUT /teachers/{id} Update teacher details
DELETE /teachers/{id} Remove a teacher


POST /teachers Add a new teacher
Parameters
Cancel
Reset
No parameters
Request body
application/json
Execute
Clear
Responses
Curl





GITHUB LINK : https://github.com/2303A52387/AI-ASSISTANT-CODING/blob/main/15.zip